

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1718

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1HR

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I **SANJAI S(192511018)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SANJAI S

CASE STUDY 3

AGRICULTURAL EXPERT REASONING SYSTEM

Given Knowledge Base (KB)

P1: Soil Dry $\rightarrow$ Irrigation Needed

P2: Irrigation Needed $\wedge$ Crop Wheat $\rightarrow$ Apply Drip Method

P3: $\neg$ Apply Drip Method $\wedge$ Crop Wheat $\rightarrow$ Crop At Risk

Facts:

- Soil Dry = True
- Crop Wheat = True

(A) CONVERT INTO CONJUNCTIVE NORMAL FORM (CNF)

P1

Original

SoilDry $\rightarrow$ IrrigationNeeded

Step 1: Eliminate implication

$\neg$ SoilDry $\vee$ IrrigationNeeded

Step 2: Move NOT inward

No change

Step 3: Distribute

Already in CNF

CNF Clause C1

$\neg$ SoilDry $\vee$ IrrigationNeeded

P2

Original

$(\text{Irrigation Needed} \wedge \text{Crop Wheat}) \rightarrow \text{Apply Drip Method}$

Step 1: Remove implication

$\neg(\text{Irrigation Needed} \wedge \text{Crop Wheat}) \vee \text{Apply Drip Method}$

Step 2: Move NOT inward

$\neg\text{Irrigation Needed} \vee \neg\text{Crop Wheat} \vee \text{Apply Drip Method}$

Step 3: Distribution

Already CNF

CNF Clause C2

$\neg\text{Irrigation Needed} \vee \neg\text{Crop Wheat} \vee \text{Apply Drip Method}$

P3

Original

$(\neg\text{Apply Drip Method} \wedge \text{Crop Wheat}) \rightarrow \text{Crop At Risk}$

Step 1

$\neg(\neg\text{Apply Drip Method} \wedge \text{Crop Wheat}) \vee \text{Crop At Risk}$

Step 2

$\text{Apply Drip Method} \vee \neg\text{Crop Wheat} \vee \text{Crop At Risk}$

Step 3

Already CNF

CNF Clause C3

$\text{Apply Drip Method} \vee \neg\text{Crop Wheat} \vee \text{Crop At Risk}$

Facts

SoilDry

CropWheat

C4: SoilDry

C5: CropWheat

Final CNF Clauses

C1 : $\neg$ SoilDry $\vee$ IrrigationNeeded

C2 : $\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ APPLYDRIPMETHOD

C3 : ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk

C4 : SoilDry

C5 : CropWheat

(B) RESOLUTION REFUTATION

Goal

Prove

ApplyDripMethod

Negate Goal

C6 : $\neg$ ApplyDripMethod

Resolution 1

C1

$\neg$ SoilDry $\vee$ IrrigationNeeded

C4

SoilDry

Resolve

C7 : IrrigationNeeded

Resolution 2

C2

$\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod

C7

IrrigationNeeded

Resolve

C8 : $\neg$ CropWheat $\vee$ ApplyDripMethod

Resolution 3

C8

$\neg$ CropWheat $\vee$ ApplyDripMethod

C5

CropWheat

Resolve

C9 : ApplyDripMethod

Resolution 4

C9

ApplyDripMethod

C6

$\neg$ ApplyDripMethod

Resolve

$\square$ (Empty Clause)

Resolution Proof Tree

$\neg$ Soil Dry $\vee$ IrrigationNeeded

|

Soil Dry

|

Irrigation Needed

|

$\neg$ Irrigation Needed $\vee$ $\neg$ Crop Wheat $\vee$ Apply Drip Method

|

Crop Wheat

|

Apply Drip Method

|

$\neg$ Apply Drip Method

|

$\square$

Conclusion

The empty clause is obtained.

Therefore,

ApplyDripMethod is proved.

(C) REMOVE SOILD DRY

Now remove

SoilDry

Remaining KB

C1

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

C2

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

C3

$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Fact

CropWheat

Step-by-Step Analysis

Without SoilDry,

Cannot infer

IrrigationNeeded

Without IrrigationNeeded,

Cannot derive

ApplyDripMethod

Without proving

ApplyDripMethod

We also cannot conclude

CropAtRisk

because the condition

$\neg \text{ApplyDripMethod}$

is not proven.

The system simply lacks enough information.

Final Conclusion

Without SoilDry,

✓ IrrigationNeeded cannot be inferred.

✓ ApplyDripMethod cannot be proved.

✓ CropAtRisk also does not follow.

Hence the reasoning chain stops.

CASE STUDY 4

WUMPUS WORLD KNOWLEDGE AGENT

Given Percepts

Stench(1,2)

Breeze(1,1)

Glitter(2,2)

Rules

R1

Stench(x,y)

→

Adjacent Wumpus

R2

Breeze(x,y)

→

Adjacent Pit

R3

Glitter(x,y)

→

Gold(x,y)

R4

$\neg$ Wumpus(x,y)

$\wedge$

$\neg$ Pit(x,y)

→

Safe(x,y)

(A) BELIEF STATE AND MODUS PONENS

Initial Belief State

Stench(1,2)

Breeze(1,1)

Glitter(2,2)

Apply Rule R1

Stench(1,2)

↓

Possible Wumpus adjacent to (1,2)

Adjacent cells

(1,1)

(1,3)

(2,2)

Apply Rule R2

Breeze(1,1)

↓

Possible Pit adjacent to (1,1)

Adjacent cells

(1,2)

(2,1)

Apply Rule R3

Glitter(2,2)

↓

Gold(2,2)

Derived Knowledge

- Possible Wumpus at (1,1), (1,3), or (2,2)
- Possible Pit at (1,2) or (2,1)
- Gold located at (2,2)

(b) Forward Chaining

Step 1

Fact

Stench(1,2)

↓

Rule R1

↓

Possible Wumpus

(1,1)

(1,3)

(2,2)

Step 2

Fact

Breeze(1,1)

↓

Rule R2

↓

Possible Pit

(1,2)

(2,1)

Step 3

Fact

Glitter(2,2)

↓

Rule R3

↓

Gold at (2,2)

Final Forward Chaining Results

Possible Wumpus

(1,1)

(1,3)

(2,2)

Possible Pit

(1,2)

(2,1)

Gold

(2,2)

(C) SAFETY ANALYSIS

Path

(1,1)

↓

(1,2)

↓

(2,2)

Cell (1,2)

May contain Pit

Not Safe

Cell (2,2)

May contain Wumpus

Although Gold exists,

Safety is not guaranteed.

Apply Rule R4

Safe(x,y)

requires

No Wumpus

AND

No Pit

Since neither is proved,

Safe cannot be inferred.

Final Conclusion

The path

$(1,1) \rightarrow (1,2) \rightarrow (2,2)$

is NOT guaranteed to be safe.

Reason:

- **Possible Pit at (1,2)**
- **Possible Wumpus at (2,2)**
- **Therefore, the agent should not directly follow this path. It must first gather additional percepts or explore alternative safe cells before attempting to collect the gold.**

CONCLUSION

- **Case Study 3: Successfully converted all rules into CNF, proved ApplyDripMethod using Resolution Refutation (ending with the empty clause $\square$), and showed that removing SoilDry breaks the inference chain, preventing both ApplyDripMethod and CropAtRisk from being concluded.**
- **Case Study 4: Applied Modus Ponens and Forward Chaining to derive possible Wumpus and Pit locations and identified the gold location. The proposed path $(1,1) \rightarrow (1,2) \rightarrow (2,2)$ is not logically safe because hazards cannot be ruled out.**